

# llmXive Automated Review of SpatialClaw: Rethinking Action Interface for Agentic Spatial Reasoning

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** [qwen.qwen3.5-122b](#)

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong quantitative claims regarding performance gains and generalization that are not fully supported by the provided data tables.

First, the most significant discrepancy lies in the reported improvement over the primary baseline, SpaceTools. The Abstract and Introduction state that SpatialClaw outperforms SpaceTools by **+11.2 points**. However, Table 1 (`tables/ablation_action_interface.tex`) explicitly lists the average accuracy for SpaceTools (Structured Tool-Call) as **56.7%** and SpatialClaw as **59.9%**. The actual difference is **3.2 points**. The +11.2 figure corresponds to the gain over the “No-tool” baseline (53.4% to 59.9%). Attributing the +11.2 gain to SpaceTools is a factual error that significantly misrepresents the magnitude of the improvement over the state-of-the-art method.

Second, the claim in Section 3 that SpatialClaw “outperforms... across all categories” is contradicted by the data in Table 1. In the “Single-image” category, specifically for the **Omni3D** benchmark, the Structured Tool-Call baseline achieves **55.7%**, while SpatialClaw achieves **54.3%**. Similarly, in the “General video understanding” category, **CV-Bench** shows Structured Tool-Call at **73.6%** versus SpatialClaw at **72.2%**. The text should be revised to acknowledge that while the *average* performance is superior, there are specific benchmarks where the structured tool-call approach remains competitive or superior.

Finally, the claim of being “training-free” and requiring “no benchmark-specific adaptation” is potentially overstated given the “Planner” description in the Appendix. The planner is described as mapping question shapes to specific tools (e.g., “coordinates -> vlm.locate + SAM3”). If this mapping logic is a fixed heuristic defined by the authors rather than a learned or fully zero-shot inference by the LLM, it constitutes a form of benchmark-specific engineering. The authors should clarify whether the planner’s logic is static or dynamically inferred to ensure the “no adaptation” claim is accurate.

**Action items.**

- **[science]** In Section 3 (Results), the text claims a +11.2 point gain over SpaceTools (Tab. 1). However, Table 1 (`action_interface`) shows SpaceTools (Structured Tool-Call) averaging 56.7% and SpatialClaw 59.9%, a difference of 3.2 points. The +11.2 figure appears to be the gain over the ‘No-tool’ baseline (53.4% -> 59.9%) or a different comparison not explicitly defined in the text. This misattribution of the baseline undermines the claim’s accuracy.
- **[writing]** Table 1 (`action_interface`) lists Omni3D results where Structured Tool-Call (55.7) outperforms SpatialClaw (54.3), yet the text in Section 3 states SpatialClaw ‘outperforms... across all categories.’ This is factually incorrect for the Omni3D benchmark within the Single-image category. The claim requires qualification or correction to reflect that gains are not universal across every single benchmark.
- **[science]** The abstract and introduction claim the framework is ‘training-free’ and achieves gains ‘without any benchmark- or model-specific adaptation.’ However, the prompt details (App. Prompts) describe a ‘Planner’ that maps question shapes to specific tools (e.g., ‘coordinates -> vlm.locate + SAM3’). If this mapping logic is hard-coded or requires manual tuning per benchmark type, the claim of ‘no adaptation’ is overstated. Clarify if the planner

is fully zero-shot or relies on pre-defined heuristics.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_figure\_critic.md

### Figure 1

The figure provides a clear and readable schematic of the agentic loop described in the caption. However, the caption contains a grammatical error in the opening sentence, and the variable  $N_{max}$  is used in the diagram without a local definition.

### Figure 2

Figure 2 effectively illustrates the three action interfaces with clear visual examples and code snippets. However, the caption text is grammatically incomplete, omitting the subject name, and does not define specific library acronyms used in the code.

### Figure 3

The figure effectively visualizes the win/loss margins across categories, but the caption contains a missing subject and incorrectly swaps the descriptions for panels (a) and (b) compared to the figure labels. Additionally, the x-axis orientation in panel (a) is non-standard.

#### Action items.

- **[writing]** Figure 1: The caption begins with a sentence fragment ('wraps a persistent kernel...') lacking a subject; the figure itself labels the system as 'SpatialClaw' (inferred from paper title) or the caption should explicitly name the method.
- **[writing]** Figure 1: The iteration condition at the bottom uses the variable ' $N_{max}$ ' which is not defined in the figure labels; while the caption mentions it, the figure text should ideally define the variable for standalone clarity.
- **[writing]** Figure 2: The caption text is incomplete and grammatically incorrect, missing the method name at the start of sentences (e.g., 'studies code' instead of 'SpatialClaw studies code', 'writes Python' instead of 'SpatialClaw writes Python').
- **[writing]** Figure 2: The caption fails to explicitly define the acronyms 'SAM3' and 'KDTree' used in the code snippets within panels (a) and (c), which may be unclear to readers unfamiliar with the specific libraries.

- **[writing]** Figure 3: The caption contains a grammatical error and missing subject in the first sentence (‘Pairwise win/loss margin of over baselines’ should be ‘margin of SpatialClaw over baselines’).
- **[writing]** Figure 3: The caption text ‘outperforms both (a) Structured tool-call and (b) Single-pass Code’ contradicts the figure labels, where (a) is ‘Single-pass Code’ and (b) is ‘Structured tool-call’.
- **[science]** Figure 3: The x-axis labels for the left panel (a) are inverted relative to the top header; the header indicates ‘Single-pass code wins’ are to the left, but the axis numbers increase to the left (10, 0, 10, 20, 30), which is non-standard and potentially confusing.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_jargon\_police.md

The manuscript relies heavily on domain-specific jargon that creates a barrier for non-specialist readers, particularly in the Introduction and Method sections.

First, the term **“agentic”** is used repeatedly (e.g., “Agentic Spatial Reasoning”, “agentic loop”) without definition. It is a buzzword in the field but means little to a general reader. The authors should use “autonomous” or “agent-based” and define the concept of an “agent” simply as a system that can make decisions and take actions.

Second, the phrase **“action interface”** is central to the paper’s contribution but is never explicitly defined in plain language. It is treated as a known concept. The authors should clarify this early on, perhaps as “the specific method or language (in this case, code) the AI uses to request help from external tools.”

Third, technical terms like **“orchestration space”** (Section 1) and **“perception primitives”** (Section 3.1) are unnecessarily dense. “Orchestration space” could simply be “workspace” or “execution environment,” and “perception primitives” could be “basic perception tools” or “pre-defined functions.” These terms add cognitive load without adding precision.

Fourth, acronyms and specific technical jargon appear without definition. **“SE(3)”** (Section 4.1) is standard in robotics but obscure to general AI researchers; it should be introduced as “3D rigid body transformations (SE(3))”. Similarly, **“AST checker”** (Section 3.2) and **“KV reuse”** (Appendix) use acronyms (Abstract Syntax Tree, Key-Value) that should be spelled out or replaced with plain English descriptions (e.g., “code syntax checker,” “caching of model states”).

Finally, the term **“backbone”** is used to refer to the underlying VLMs (e.g., “Qwen3.5-397B backbone”). While common in the field, “base model” or “underlying model” is more accessible.

The paper’s core idea—using a persistent code environment for spatial reasoning—is clear, but the presentation is cluttered with jargon that excludes readers outside the specific sub-field

of agentic systems and robotics. A revision to simplify these terms would significantly improve the paper’s accessibility.

#### Action items.

- **[writing]** The manuscript relies heavily on domain-specific jargon that creates a barrier for non-specialist readers, particularly in the Introduction and Method sections. First, the term “agentic” is used repeatedly (e.g., “Agentic Spatial Reasoning”, “agentic loop”) without definition. It is a buzzword in the field but means little to a general reader. The authors should use “autonomous” or “agent-based” and define the concept of an “agent” simply as a system that can make decisions and take actions. Sec

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_logical\_consistency.md

The logical consistency of the paper is generally strong regarding the core mechanism: the persistent kernel allows for iterative refinement, which logically explains the gains on complex, multi-step tasks (e.g., Video/4D reasoning). The causal chain from “iterative code execution” to “improved spatial reasoning” is well-supported by the ablation studies in Table 3, which isolate the contribution of the interface from the tools.

However, there are specific instances where the textual claims overreach the data presented, creating minor logical gaps:

1. **Overgeneralization of “Consistent Gains”:** The abstract and conclusion state that gains are “consistent across six VLM backbones.” Table 1 (tab:main\_results) explicitly shows negative deltas for the Qwen3.5-397B model on ERQA (-1.0) and Omni3D (-1.9). While the *average* is positive, the word “consistent” implies a uniform direction of improvement which the data refutes. This should be rephrased to “consistent average gains” or “gains on the majority of benchmarks.”
2. **Ablation Interpretation:** In Section 5, the text claims that removing utility functions yields performance “on par” with the full system. Table 3 (tab:ablation) shows the “No utility” variant (56.4%) is strictly lower than the “Full” variant (56.9%) and underperforms on 12 of the 20 benchmarks. While the difference is small (0.5%), describing it as “on par” is logically imprecise when the full system is strictly superior. The text should acknowledge the marginal but consistent degradation to maintain logical rigor.
3. **Causal Attribution:** The paper attributes performance entirely to the “action interface.” However, the results show that the magnitude of improvement varies drastically by backbone (e.g., +18.6% for Qwen3.5-122B on Omni3D vs +0.7% for Qwen3.5-397B). The logical link between the interface and the *magnitude* of the gain is not fully explored. It is possible that smaller models benefit more from the structured code interface than larger, more capable

models that might already reason well via chain-of-thought. The discussion should address whether the interface is a substitute for model capability or a multiplier, to avoid a simplistic causal claim.

These issues are primarily matters of precise wording and nuance rather than fundamental flaws in the experimental design or the core hypothesis.

#### Action items.

- **[writing]** The claim that ‘Removing utility functions (I) yields performance on par with full SpatialClaw’ (Sec 5) is contradicted by Table 3. The ‘No utility’ variant (56.4%) is 0.5% lower than ‘Full’ (56.9%) and underperforms on 12/20 benchmarks. Clarify that the drop is marginal rather than ‘on par’ to avoid overstatement.
- **[writing]** Table 1 shows Qwen3.5-397B performance decreases on ERQA (-1.0) and Omni3D (-1.9) with SpatialClaw. However, the abstract claims ‘consistent gains across six VLM backbones’. This is a logical contradiction; qualify the claim to ‘consistent average gains’ or ‘gains on most benchmarks’.
- **[science]** The paper claims SpatialClaw is ‘training-free’ and attributes gains solely to the interface. However, Table 1 shows massive variance in gains across backbones (e.g., +18.6% vs +0.7% on Omni3D). The text must discuss whether the interface benefits smaller models more or if backbone capabilities confound the causal claim, rather than attributing all variance to the interface.

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper makes several strong claims regarding the universality and mechanism of its performance gains that exceed the evidence provided in the tables and ablation studies.

First, the Abstract and Introduction assert that the framework achieves “consistent gains across six VLM backbones.” This phrasing implies a uniform positive delta across all models and tasks. However, Table 1 (main\_results.tex) explicitly shows that for the Qwen3.5-397B backbone, the method yields negative deltas on four single-image benchmarks (ERQA: -1.0, Omni3D: -1.9, ViewSpatial: -5.1, Video-MME: -0.5). While the *average* is positive, claiming “consistent gains” without qualification misrepresents the data, suggesting a robustness that the specific per-benchmark results do not support. The text should be amended to reflect that gains are consistent *on average* or *across most benchmarks*, rather than universally.

Second, the claim of operating “without any benchmark- or model-specific adaptation” (Abstract) is technically inaccurate. Section 3.2 and Appendix D describe a “Planner” role that is explicitly invoked to “map question shapes to tools” (e.g., coordinates -> vlm.locate + SAM3). This is a form of prompt-based adaptation or heuristic routing specific to the task

structure. While the authors likely mean “without fine-tuning,” the current phrasing suggests a zero-shot, zero-prompt-tuning capability that the described system does not possess. This distinction is crucial for reproducibility and fair comparison with baselines that may also use prompt engineering.

Finally, the Conclusion attributes the success primarily to the “action interface design” as a “critical driver.” The ablation study in Table 2 (ablation\_components.tex) challenges this interpretation. Variant (I) “No utility functions” (removing the specific code wrappers) only drops performance by 0.5% (56.9% to 56.4%), suggesting the *interface* (code execution) is indeed important, but the *specific utility functions* are not the primary driver. More critically, Variant (II) “No perception tools” drops performance to 51.4%, which is only a 2.7% gain over the No-tool baseline (48.7%). This indicates that the bulk of the performance (from 48.7% to 56.9%) is driven by the *perception tools* (Depth Anything 3, SAM3) rather than the iterative code interface itself. The paper over-attributes the success to the interface mechanism while under-acknowledging that the gains are largely contingent on the quality of the external perception models. The discussion should be balanced to reflect that the interface enables the *composition* of these tools, but the tools themselves are the primary source of the absolute performance lift.

#### Action items.

- **[writing]** The claim of ‘consistent gains’ across all backbones (Abstract) contradicts Table 1 (main\_results.tex), where Qwen3.5-397B shows negative deltas on 4/10 single-image benchmarks (e.g., -1.0 on ERQA). Qualify as ‘consistent on average’.
- **[writing]** The claim of ‘no benchmark-specific adaptation’ (Abstract) is contradicted by the ‘Planner’ role (Sec 3.2, App D) which maps question shapes to tools. Refine to ‘without fine-tuning’ to avoid over-claiming prompt engineering as zero-shot.
- **[science]** The conclusion over-attributes success to the ‘action interface’ (Sec 6). Ablation (Table 2) shows removing perception tools drops performance to 51.4% (vs 48.7% No-tool), indicating tools drive most gains, not just the interface. Balance the claim.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_safety\_ethics.md

The manuscript presents SpatialClaw, a framework for agentic spatial reasoning using a persistent Python kernel. From a safety and ethics perspective, the paper identifies the core mechanism (code execution) as a strength but lacks sufficient depth in its risk assessment.

**Dual-Use and Malicious Application:** The “Broader Impact” section (Section app:broader\_impact) is currently too generic, stating only that the system is “valuable for robotics and assistive systems.” Given that the framework allows an LLM to write and execute arbitrary Python code to control perception tools (segmentation, 3D reconstruction) and poten-



tially interact with external hardware (via the “robotics” mention), the potential for dual-use is significant. The authors must expand this section to explicitly discuss risks such as:

1. **Autonomous Surveillance:** The ability to chain segmentation and tracking tools could be used to build automated surveillance systems that violate privacy norms.
2. **Physical Security Breaches:** If integrated with robotic actuators (implied by the “agentic” nature), the system could be manipulated to bypass physical security measures or cause physical harm.
3. **Automated Exploitation:** The code-generation capability could be repurposed to generate exploits for spatial vulnerabilities in software or hardware systems.

**Security of the Execution Environment:** The “Security Sandbox” (Section app:sandbox) describes a defense-in-depth strategy using AST traversal and regex to block unsafe modules (e.g., `os`, `socket`, `eval`). However, the review notes a lack of discussion regarding the *threat model* of the agent itself. An advanced LLM agent might attempt to:

1. **Jailbreak the Sandbox:** Craft code that bypasses static analysis (e.g., using obfuscation or dynamic attribute access) to access restricted resources.
2. **Exploit Tool Vulnerabilities:** The framework relies on external perception tools (Depth Anything 3, SAM3). If these tools have vulnerabilities, the agent could potentially trigger them to execute arbitrary code or leak memory, effectively bypassing the sandbox.

The authors should clarify whether the sandbox is designed to be robust against adversarial agent behavior or if it assumes a benign agent. A brief discussion on the limitations of static analysis in this context is required.

**Data Privacy and Consent:** The system processes images and videos (up to 32 frames per sample) through a persistent kernel. While the paper mentions “open-source backbones,” it does not address the privacy implications of processing real-world data. If this framework is deployed in applications involving human subjects (e.g., assistive robotics, autonomous driving), there are significant concerns regarding:

1. **PII Exposure:** The system might inadvertently process or store Personally Identifiable Information (PII) present in the input images.
2. **Consent:** There is no mention of how consent is obtained for data processing, especially if the system is used in public spaces.

The authors should add a paragraph in the “Limitations” or “Broader Impact” section addressing data privacy, retention policies, and the necessity of user consent in real-world deployments.

**Conclusion:** While the technical contribution is sound, the safety and ethics discussion is currently superficial. The paper requires a more rigorous treatment of dual-use risks, the security limitations of the proposed sandbox, and data privacy considerations before it can be considered fully responsible.

**Action items.**



- **[writing]** Jailbreak the Sandbox: Craft code that bypasses static analysis (e.g., using obfuscation or dynamic attribute access) to access restricted resources.
- **[writing]** Exploit Tool Vulnerabilities: The framework relies on external perception tools (Depth Anything 3, SAM3). If these tools have vulnerabilities, the agent could potentially trigger them to execute arbitrary code or leak memory, effectively bypassing the sandbox. The authors should clarify whether the sandbox is designed to be robust against adversarial agent behavior or if it assumes a benign agent. A brief discussion on the limitations of static analysis in this context is required. Data Privacy
- **[writing]** PII Exposure: The system might inadvertently process or store Personally Identifiable Information (PII) present in the input images.
- **[writing]** Consent: There is no mention of how consent is obtained for data processing, especially if the system is used in public spaces. The authors should add a paragraph in the “Limitations” or “Broader Impact” section addressing data privacy, retention policies, and the necessity of user consent in real-world deployments. Conclusion: While the technical contribution is sound, the safety and ethics discussion is currently superficial. The paper requires a more rigorous treatment of dual-use risks, the

## Scientific Evidence

*Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling architectural shift for spatial reasoning agents, but the scientific evidence supporting the magnitude and robustness of the claims requires strengthening. While the average gain of +11.2 points over SpaceTools is substantial, the manuscript currently lacks statistical rigor to confirm these are not artifacts of specific benchmark splits or random initialization variance.

First, the results sections (Sec. 4, Tab. 1, Tab. 2) report only point estimates (e.g., 59.9% vs 48.7%). Without reported standard deviations, confidence intervals, or p-values derived from multiple runs or bootstrap resampling, it is impossible to assess the statistical significance of the gains. For instance, the +17.6 point gain on DSI-Bench (Tab. 2) is a massive effect size, but without variance data, we cannot rule out that this specific benchmark’s distribution was unusually favorable to the proposed method.

Second, the sample size per benchmark is ambiguous. The Appendix mentions a “1,000 samples max” limit, but it is unclear if every benchmark utilized exactly 1,000 samples or if the counts varied. If sample sizes differ significantly across the 20 benchmarks, the simple arithmetic mean (59.9%) is a biased estimator of the true performance. The authors must explicitly state the N for each benchmark and, if N varies, justify the averaging method or provide a weighted average.

Third, the ablation study (Tab. 4) claims that removing utility functions or perception tools yields specific performance drops. However, without error bars or significance testing, the small differences (e.g., 56.9 vs 56.4 average) could be within the noise floor of the evaluation. The claim that “composition is the main driver” relies on these marginal differences; robust evidence requires demonstrating that these drops are statistically significant.

Finally, the evaluation relies on a single run per configuration for the main tables. Given the stochastic nature of LLM generation and the complexity of the code execution loop, a single run is insufficient to establish the reliability of the reported scores. The authors should report results averaged over at least 3-5 independent runs with seeds, or provide a rigorous analysis of the variance across the 20 benchmarks to support the “consistent gains” claim.

#### Action items.

- **[science]** Report statistical significance (p-values or confidence intervals) for the reported +11.2 point average gain and key benchmark improvements (e.g., DSI-Bench +17.6) to rule out random variance, as only point estimates are currently provided.
- **[science]** Clarify the sample size (N) per benchmark. The text mentions ‘1,000 samples max’ (App. E) but does not confirm if all 20 benchmarks used the full N or if some were subsampled, which affects the validity of the aggregate average.
- **[science]** Provide a variance measure (standard deviation or standard error) for the ablation study results (Tab. 4) to demonstrate that the observed differences between ‘Full’, ‘No utility’, and ‘No perception’ are robust and not due to noise.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_statistical\_analysis.md

The manuscript presents a compelling architectural shift for spatial reasoning agents, but the statistical rigor supporting the quantitative claims requires strengthening. While the performance gains are substantial (e.g., +11.2% average accuracy), the analysis lacks formal statistical validation.

First, the paper reports point estimates for accuracy across 20 benchmarks (e.g., Table 1, Table 2) but does not provide measures of uncertainty. For a claim of “consistent gains,” it is essential to report confidence intervals (e.g., 95% CI) or standard errors for the mean accuracy. Without these, it is impossible to determine if the observed improvements are statistically significant or potentially due to random variance in the benchmark samples. Specifically, the claim that the method outperforms SpaceTools by +11.2 points (Section 1, Abstract) needs a p-value or a significance test (e.g., paired t-test or Wilcoxon signed-rank test across benchmarks) to support the assertion of a robust improvement.

Second, the evaluation involves a large number of comparisons: 20 benchmarks across 6

different model backbones. The paper does not mention any correction for multiple comparisons. In a setting with 120 total comparisons (20 benchmarks  $\times$  6 models), the likelihood of Type I errors (false positives) is non-negligible. The authors should explicitly state whether they applied corrections (such as Bonferroni or Holm-Bonferroni) or if the results are presented as exploratory. If no correction was applied, the interpretation of “consistent gains” should be tempered to reflect the increased risk of false discoveries.

Third, the definition of the “Average” metric in Table 1 and Table 2 needs clarification. Is this a simple arithmetic mean of the benchmark accuracies, or is it weighted by the number of samples in each benchmark? Given that benchmarks vary significantly in difficulty and sample size (e.g., Video-MME vs. SPBench), a simple mean might be biased. The authors should specify the aggregation method and justify why it is appropriate for comparing performance across such diverse tasks.

Finally, while the ablation studies (Table 3) show performance drops when components are removed, the statistical significance of these drops is not quantified. Are the differences between the “Full” model and the “No utility” or “No perception” variants statistically significant? Providing p-values for these ablation results would strengthen the conclusion that the specific components are critical drivers of performance.

In summary, the paper’s conclusions are promising, but the statistical evidence is currently insufficient to fully support the strength of the claims. Adding significance testing, confidence intervals, and clarifying the aggregation and multiple-comparison strategies is necessary.

#### Action items.

- **[science]** Report statistical significance (e.g., p-values, confidence intervals, or effect sizes) for the reported average accuracy gains (+11.2 points) and per-benchmark improvements. The current text claims consistent gains but lacks formal hypothesis testing to rule out random variance across the 20 benchmarks.
- **[science]** Clarify the multiple-comparisons handling strategy. With 20 benchmarks and 6 backbones, the probability of false positives increases. Explicitly state if corrections (e.g., Bonferroni, Holm-Bonferroni) were applied or if the analysis is treated as exploratory.
- **[science]** Define the unit of analysis for the reported averages. Specify whether the ‘Average’ metric in Table 1 and Table 2 is a simple arithmetic mean of benchmark accuracies or a weighted mean based on sample counts, and justify the choice given the varying difficulty and size of the 20 benchmarks.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The manuscript is generally well-written with a clear narrative flow and precise technical

descriptions. The introduction effectively sets up the problem of action interfaces in spatial reasoning, and the methodology is described with sufficient detail for reproducibility. However, there are a few specific writing and formatting issues that need correction before publication.

First, in Section 5 (“Results”), the authors use a custom command `\paragrapht` to introduce paragraphs. This command is not defined in the provided LaTeX preamble, which will result in a compilation error. The authors should either define this command or replace it with the standard `\paragraph` environment to ensure the document compiles correctly.

Second, there is a broken cross-reference in Section 5. The text states: “outperforms Single-Pass Code and Structured Tool-Call (SpaceTools~\citep{chen2025spacetools}) by +11.2 points on average (Tab.~\ref{tab:comparison\_agents}).” However, the label `tab:comparison_agents` does not exist in the document. The data described corresponds to `tab:action_interface`. This reference must be updated to point to the correct table label.

Third, the caption and content for Figure 2 (`method_loop`) appear to be duplicated. The figure and its caption are present in the main body (Section 3) and again in the Appendix (Section “Additional Analysis”). This duplication will likely cause numbering conflicts or redundant content in the final PDF. One instance should be removed or converted to a reference.

Finally, while the writing is strong, the consistency of terminology could be slightly improved. For instance, the paper alternates between “Single-pass code” and “Single-Pass Code” in table headers and text. Standardizing this capitalization would enhance professional polish. Overall, these are minor mechanical issues that do not detract from the clarity of the scientific contribution but should be addressed for a clean final version.

#### Action items.

- **[writing]** In Section 5 (Results), the custom command `\paragrapht` is used but not defined in the preamble. This will cause a compilation error. Replace with standard `\paragraph` or define the command.
- **[writing]** In Section 5, the text references ‘Tab.~\ref{tab:comparison\_agents}’, but no such label exists in the document. The intended reference is likely ‘Tab.~\ref{tab:action\_interface}’.
- **[writing]** The caption for Figure 2 (`method_loop`) appears twice in the source (once in the main body, once in the appendix). Ensure only one instance remains to avoid duplicate figure numbering.
- **[writing]** In the ‘Backbones’ section (Appendix), the table caption refers to ‘Table~\ref{tab:backbones}’, but the label is defined inside the table environment. While often tolerated, ensure the label is placed immediately after `\caption` to guarantee correct cross-referencing.